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SPECIAL ISSUE ORIGINAL STUDY

The Role of Geographic Information Systems in Promoting the 15-minute City Concept for Sustainable Urban Using Network Analysis: A Case Study: El Rehab City

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Abstract

The 15-minute city concept promotes urban accessibility and sustainability by ensuring essential services are available within a 15 min walk or bike ride. This study examines the feasibility of this model in El Rehab City, Egypt, using geographic information systems (GIS) and network analysis to evaluate the accessibility of key services such as healthcare, education, and recreation. Through data modeling of the road network, pedestrian paths, and bike lanes, and by calculating shortest paths, travel times, and accessibility metrics, the study identifies strengths, such as 85 % service coverage within a 15 min cycling distance, and weaknesses, including the limited coverage of medical services at only 19 % within walking range. These results underscore critical gaps in urban equity and highlight the need for decentralized service hubs and enhanced public transportation. By leveraging GIS-based methodologies, the research offers actionable insights to improve service accessibility, reduce car dependency, and foster sustainable urban development. This research underscores GIS's transformative role in urban planning, demonstrating how data-driven methodologies can inform sustainable, inclusive, and resilient urban environments while advancing the global discourse on the 15-minute city model.

Keywords: El Rehab City, Geographic information systems (GIS), Network analysis, Planned cities, Sustainability, The 15-minute city, Urban accessibility

1. Introduction

Urbanization has significantly transformed city landscapes worldwide, necessitating innovative planning strategies that prioritize urban accessibility, sustainability, and quality of life (Moreno *et al.*, 2021). Among these strategies, the 15-minute city concept has emerged as a transformative response to urban challenges such as traffic congestion, pollution, and social inequality (Allen and Farber, 2020). This model promotes the development of planned cities and urban environments where residents can meet their daily needs—work, education, healthcare, and recreation—within a 15 min walk or bike ride. By emphasizing sustainability and proximity, this approach seeks to reduce car dependence (Horton, 2024), lower carbon emissions, encourage active mobility, and foster a sense of community by bringing essential services closer to where people live (Bright, 2021).

While the 15-minute city presents a compelling vision for sustainable urban living, its implementation is fraught with challenges (Christidis *et al.*, 2022). Globally, socioeconomic inequalities, urban sprawl, and the need for substantial infrastructure investments complicate the realization of this model

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(Allam et al., 2024). For instance, retrofitting existing cities to align with urban accessibility principles and the 15-minute city framework requires addressing entrenched urban designs and spatial inequities (Jing *et al.*, 2024). Cities like Paris and Melbourne have piloted the concept with varying degrees of success, revealing its transformative potential and inherent limitations. Critiques also highlight that the model may overlook marginalized populations and economic realities, necessitating a more inclusive approach to urban planning (Winslow and Coenen, 2023).

In Middle Eastern contexts, the 15-minute city encounters additional hurdles rooted in cultural, climatic, and urban design factors. Preferences for private cars, societal norms around public spaces, and family-oriented lifestyles challenge the adoption of walkable, mixed-use developments (Papadopoulos *et al.*, 2023). Furthermore, extreme climatic conditions, such as high temperatures, limit the feasibility of outdoor mobility for large parts of the year. The prevalence of gated communities and a lack of integrated public transport systems exacerbate these issues, restricting urban accessibility and intensifying socioeconomic disparities (Camerin, 2023). Addressing these challenges requires planned cities to adopt tailored strategies that align with regional realities while learning from global best practices (Yue *et al.*, 2024).

Geographic information systems (GIS) have become indispensable in modern urban planning, offering sophisticated tools for spatial data analysis and the optimal distribution of urban services (Ferrer-Ortiz *et al.*, 2022). Through network analysis, GIS enables a detailed evaluation of connectivity and accessibility within urban areas (Mohamed *et al.*, 2024). This research leverages these capabilities to explore the application of GIS in advancing the 15-minute city concept, using El Rehab City, a planned city in Cairo, Egypt, as a case study.

El Rehab City provides a unique context for assessing how GIS can guide urban design strategies aligned with the 15-minute city model. Its planned infrastructure offers an opportunity to evaluate the spatial distribution of essential services and their accessibility to residents. This study begins with an overview of the 15-minute city concept and its core principles. It then examines the role of GIS in urban planning, focusing on network analysis as a critical tool for evaluating urban accessibility. Additionally, the study explores global experiences with the 15-minute city, offering insights into their applicability to the Middle Eastern context.

This research evaluates the feasibility of implementing the 15-minute city concept in El Rehab City, Egypt, by assessing the accessibility of essential urban services, including healthcare, education, recreation, and commercial facilities, within a 15 min walking or cycling distance. The study aims to evaluate the city's urban infrastructure in terms of accessibility and equity, identifying strengths and weaknesses, and proposing actionable strategies to enhance sustainability and inclusivity. Using GIS-based network analysis, the research models and analyzes the spatial distribution of services. The methodology includes mapping the city's road network, pedestrian paths, and bike lanes, calculating travel times and shortest paths to services, and quantifying service coverage to identify underserved areas. The study seeks to answer three key research questions: (1) How accessible are essential services in El Rehab City within a 15 min walking or cycling distance? (2) What are the strengths and weaknesses of the city's infrastructure in supporting the 15-minute city model? (3) What policy interventions are required to improve service accessibility and promote sustainable urban development? The research underscores the potential of GIS-based methodologies to guide urban planning practices toward sustainable, equitable, and resilient urban environments. Ultimately, the findings contribute to the global discourse on the 15-minute city model, offering insights that can inform urban planning in other cities facing similar challenges.

Abbreviations

GIS	geographic information systems
PPPs	public-private partnerships
SWOT	Strengths, Weaknesses, Opportunities, and Threats
TMG	Talaat Mostafa group

2. Literature review

2.1. The 15-minute city: a concept for sustainable urban planning

2.1.1. The 15-minute city concept

The 15-Minute City is an innovative urban planning concept that designed to establish compact, self-sufficient neighborhoods where residents can access essential services and amenities within a 15 min walk or bike ride from their homes (Thompson and Inclusivity, 2024). Developed by urban planning scientist Carlos Moreno in 2016 (Poorthuis and Zook, 2023), this model is based on the principles of sustainability, accessibility, and proximity, with the goal of reducing car dependency, minimizing environmental impact, and enhancing the quality of life for urban residents (Sepe, 2022).

The 15-minute city timeframe is designed to balance practicality and accessibility, enabling residents to reach essential services within a 15 min walk or bike ride. Covering about 1.2 km (¾ mile), this distance is comfortable for most people and aligns with public health recommendations promoting moderate physical activity (Koljensic, 2022). Shorter timeframes, such as 5 or 10 min, would limit access to key services, while longer ones could encourage car use due to the increased distance. The 15-min model also supports sustainability, reducing car reliance and encouraging healthier lifestyles by promoting walking and cycling (Xu, 2021).

2.1.2. Key principles, benefits

The key components of the 15-minute city—accessibility, mixed-use development, sustainable transportation, green spaces, and community engagement—are interrelated and collectively contribute to creating livable, localized urban environments (Jing *et al.*, 2024). Accessibility is the central pillar, ensuring that essential services such as healthcare, education, groceries, and recreation are within a 15 min walking or cycling distance (Shoina *et al.*, 2024). Mixed-use development supports this by integrating residential, commercial, and recreational zones into compact areas, reducing the need for long commutes (Abdelfattah *et al.*, 2022). Sustainable transportation, including cycling and walking infrastructure, ties directly to accessibility, enabling safe and efficient movement within the city (Louro *et al.*, 2021). Green spaces enhance environmental quality and promote physical and mental well-being, while community engagement fosters a sense of belonging and local identity (Akrami *et al.*, 2024).

The 15-minute city is based on several core principles as shown in Fig. 1:

(a) Proximity and accessibility: Essential services should be accessible within a 15 min walk or bike ride, reducing the need for long commutes, and encouraging active transportation (Pandey *et al.*, 2022).
(b) Mixed-use development: Integrating residential, commercial, and recreational spaces within neighborhoods fosters vibrant communities and supports local economies (Abdelfattah *et al.*, 2022).
(c) Sustainable Transportation: Prioritizing walking, cycling, and public transit over car dependency reduces traffic congestion, lowers carbon emissions, and promotes public health (Li *et al.*, 2022).
(d) Social equity: Ensuring equal access to essential services for all residents, regardless of socioeconomic status, promotes social cohesion, and inclusion (Bright, 2021).
(e) Digitalization: Incorporating technology into urban planning enhances connectivity and accessibility, facilitating remote work, and online services (Chau *et al.*, 2022).

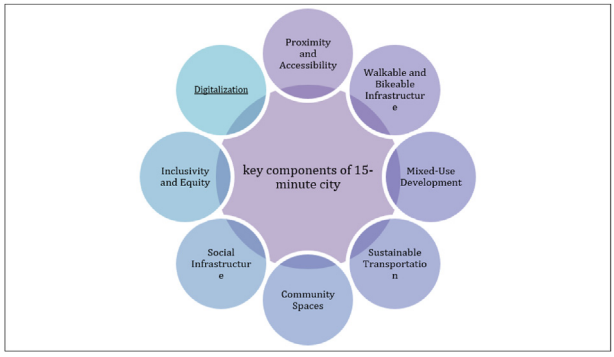


Fig. 1. The key components of a successful 15-minute city Source: by authors.

Among these components, accessibility has the strongest effect on a city's transformation because it directly determines the functionality and success of the 15-minute city model. Without accessible services and infrastructure, other components such as sustainable transportation or mixed-use development cannot achieve their full potential. Accessibility acts as the foundation, enabling all other elements to thrive and support a cohesive urban ecosystem (Shoina *et al.*, 2024).

The 15-minute city offers numerous scientific benefits. Environmentally, it contributes to climate change mitigation by reducing greenhouse gas emissions from transportation. Economically, it supports local businesses and employment opportunities (Louro *et al.*, 2021). Socially, it enhances quality of life by reducing commute times, increasing physical activity, and fostering stronger community bonds (Graells-Garrido *et al.*, 2021).

2.1.3. The main challenges in implementing the 15-minute city concept

The 15-minute city concept seeks to create urban environments where residents can access essential services within a 15 min walk or bike ride. However, several challenges hinder its effective implementation. Social and economic inequalities may prevent vulnerable populations from benefiting equally, leading to further marginalization (Allen and Farber, 2020). Additionally, many existing infrastructures may not support necessary changes, making retrofitting complex and costly, especially in sprawling cities (Rizk, 2021). Active community involvement is crucial for ensuring diverse urban

lifestyles are represented, but engaging all stakeholders can be difficult in areas with varying civic engagement levels (Tam, 2022). The transition may also create mobility concerns for those reliant on cars, requiring a balance between different transport modes (Khavarian-Garmsir *et al.*, 2023b). Furthermore, the model must be tailored to the unique characteristics of each urban area to avoid inconsistent outcomes. Lastly, efficient logistics and service delivery in a compact urban model present challenge, as maintaining sustainable supply chains becomes increasingly critical (Sanvicens, 2023).

2.2. The role of GIS and network analysis in promoting the 15-minute city concept

GIS and network analysis are critical tools for the development and implementation of the 15-minute city concept, facilitating the planning and evaluation of residents' accessibility to essential services within a 15-min walking or cycling distance (Chau *et al.*, 2022). These technologies facilitate detailed spatial analysis and visualization of urban environments, enhancing urban planning efforts. Here's how GIS and network analysis contribute to this objective (Pozoukidou and Angelidou, 2022):

(a) Creation of spatial databases: A spatial database can be built using GIS, containing up-to-date information about all available services. This facilitates residents' access to services through smartphone apps or online platforms (Wang *et al.*, 2024).
(b) Service distribution analysis: Using GIS, accurate maps can be created to show the distribution of services such as schools, hospitals, shopping centers, parks, public transport, and more. This allows for identifying how close residents are to these services and how to improve their distribution to ensure accessibility within 15 min on foot or by public transport (Khavarian-Garmsir *et al.*, 2023a).
(c) Distance and time analysis: GIS can measure distances and the time required to reach various services from any point in the city. By using data related to road networks and public transport, it is possible to identify areas where services are distant from some residents and work on reducing these distances (Akrami *et al.*, 2024).
(d) Infrastructure design and improvement: GIS helps in planning and analyzing transportation routes and urban networks, contributing to improving mobility within the city, whether by upgrading roads or enhancing public transport to ensure quick access to different services (Chabaud and Pratlong, 2024).
(e) Supporting decision-making: By analyzing spatial data, decision-makers can improve urban policies, such as allocating resources more efficiently and prioritizing urban projects that enhance the realization of the '15-minute city' concept (Khavarian-Garmsir *et al.*, 2023a).

In this research, GIS and network analysis will be used to determine the areas covered by the services within 15 min of walking or cycling by applying to El Rehab City. This is through the creation of a database that includes the road network and various service points, enabling an analysis of the network. The 'service area' tool is then used to identify service aggregation areas and determine the scope of impact of these areas.

2.3. Global experiences with the 15-minute city concept

Numerous cities worldwide have successfully implemented the 15-minute city concept, showcasing innovative urban planning practices that prioritize accessibility, sustainability, and enhanced quality of life. Paris, France, is a pioneer of this model, inspired by urban planner Carlos Moreno, with nearly 50 neighborhoods functioning as 15-min cities through investments in pedestrian zones, bike lanes, and green spaces (Hellervik *et al.*, 2019). Similarly, Ottawa, Canada, embraces the concept under the term '15-min neighborhoods,' fostering compact, connected, and complete communities to improve access to services and promote social engagement (Winslow and Coenen, 2023). Amsterdam, Netherlands, emphasizes walkable neighborhoods and sustainable transportation, leveraging cycling patterns and air quality data to ensure access to essential services and enhance residents' quality of life (Marquet and Miralles-Guasch, 2015). In New York City, USA, initiatives like bike paths, pedestrian plazas, and outdoor dining areas aim to reclaim street space for public use, prioritizing people over vehicles and fostering community interaction (Xu, 2021). These global examples highlight the transformative potential of the 15-minute city in reimagining urban living.

2.3.1. Barcelona, Spain

Barcelona is a global frontrunner in implementing the 15-minute city concept through its 'Superblocks'

initiative, launched in 2016. This approach divides the city into units known as 'Superblocks,' each covering an area of approximately 160 000 m^2 (Chang, 2019). The initiative aims to enhance quality of life by prioritizing active mobility, reducing car dependence, and fostering a sense of community.

Key features of the superblocks model:

(a) Essential services such as healthcare, education, recreation, transportation, and supplies are accessible within a 15 min walking or cycling radius (Guzman *et al.*, 2024).
(b) Internal streets are closed to regular vehicular traffic, designated instead for pedestrians and cyclists.
(c) Vehicle access is limited to residents and delivery services, with a maximum speed limit of 10 km/h.
(d) Streets within Superblocks are transformed into public squares and green spaces, encouraging community interaction and creating eco-friendly urban environments.
(e) Increased availability of small markets, medical centers, and local schools in each neighborhood (Wang *et al.*, 2024).

Challenges:

(a) Economic: The initiative requires significant financial resources to repurpose public spaces and build the necessary infrastructure to support pedestrians, cyclists, and localized services.
(b) Social: Cultural habits, such as dependence on private vehicles, and resistance from business owners concerned about restricted car access affecting their operations, pose significant obstacles (TMG, 2024).

2.3.2. *Melbourne, Australia*

Melbourne's adaptation of the concept, known as '20-min neighborhoods,' shares the goals of the 15-minute city by ensuring residents can meet daily needs within a 10 min walk (equivalent to an 800 m radius) to essential destinations, reducing reliance on cars while promoting physical activity (Abdelfattah *et al.*, 2022).

Key elements of the 20-min neighborhood model:

(a) Providing Essential Services and Facilities: Including education, healthcare, recreational amenities, and local businesses.
(b) Enhancing Infrastructure for Active Mobility: Expanding pedestrian and cycling networks while improving public transportation systems to ensure interconnectivity (Girard *et al.*, 2022).

Implementation Strategies:

(a) Establishing 'Neighborhood Activity Centers' that integrate schools, healthcare facilities, parks, cultural services, and local stores within each neighborhood.
(b) Encouraging walking and cycling through expanded infrastructure and promoting electric vehicles to reduce carbon emissions.
(c) Increasing green spaces and community areas for recreation.
(d) Supporting small and local businesses to limit the need for travel to centralized hubs.
(e) Leveraging technology for resource management, such as mobile applications that help residents locate nearby services (Oh and Dong-O'brien, 2023).

Challenges:

(a) Urban planning: Melbourne's sprawling urban structure and heavy reliance on cars make transitioning to a 15-minute city model challenging, especially in areas lacking essential services.
(b) Climatic factors: Significant weather fluctuations, including hot summers and rainy seasons, deter walking and cycling (Allen and Farber, 2020).

These global experiences illustrate the adaptability of the 15-minute city concept, highlighting diverse approaches to creating more sustainable and equitable urban environments. While these examples showcase successful implementations, they also underscore the challenges of adapting the model to different cultural, economic, and environmental contexts, tailored to the city's unique urban, cultural, and climatic contexts. Barcelona's Superblocks inspire the creation of pedestrian-centered neighborhoods by repurposing streets into green spaces and plazas while ensuring essential services are within walking or cycling distance. Melbourne's decentralized service hubs and active mobility infrastructure, adapted with shaded pathways, can enhance accessibility in hot climates. Paris emphasizes walkability through expanded green spaces and bike-sharing systems, while New York City's reclaimed street spaces offer a model for transforming car-dominated areas into community-centered public spaces.

Building on these experiences, a framework can be established to identify the essential components required for implementing the 15-minute city concept. The process of implementation unfolds across several key stages:

(a) Access to essential services: Most of the mentioned models focus on providing essential services such as healthcare, education, local markets, and recreational facilities within a short distance (15 min on foot or by bike).
(b) Promoting active mobility and reducing car dependence: Investing in infrastructure for walking and cycling, such as pedestrian pathways and protected bike lanes.

Restricting traffic in residential areas and reducing reliance on private vehicles in favor of public transportation, walking, and cycling.

(c) Increasing green spaces and reallocating land use: Creating green spaces and public squares within neighborhoods to encourage community interaction and improve the urban environment. Transforming streets into public spaces, such as plazas and open areas for community use.
(d) Supporting the local economy: Enhancing the presence of local shops and small businesses to meet residents' needs and minimize the need for long-distance travel.
(e) Utilizing smart technology: Leveraging data and technology (such as GIS and various mobile applications) to optimize service distribution and area planning.
(f) Addressing climatic challenges: Designing infrastructure to adapt to local climatic conditions, such as providing shaded pathways in hot regions.
(g) Community engagement: Encouraging community participation and collaboration with local stakeholders to ensure the concept is well-received and effectively implemented.

3. Material and methods

3.1. Case study: El Rehab City (A model for the 15-minute city concept)

El Rehab City, located in New Cairo, Egypt, is a pioneering example of a privately developed, large-scale residential community by Talaat Mostafa group in the late 1990s, planning as shown as Fig. 2. Al Rehab City covers a total area of 10 million m^2 and is designed to house 300 000 residents within 40 000 residential units (Yue *et al.*, 2024). The city has embraced a clear and innovative vision for urban growth since its inception, as reflected in its slogan, 'come and live in Al-Rehab.' This slogan served as a guiding vision for officials and staff, motivating them to create and deliver a highly livable community (Staricco and Brovarone, 2022). Designed as a self-sufficient, integrated urban environment, El Rehab embodies key principles of modern urban planning as shown as Table 1, offering residents access to a comprehensive range of services and amenities.

El Rehab City is an ideal candidate for exploring the 15-minute city concept due to its planned urban design, which features distinct residential, commercial, and green zones, promoting easy access to essential services within walking or cycling distance, as shown as Fig. 3. The city's mid-sized layout offers scalability, making it possible to apply insights from its development to other similar cities in Egypt. Despite its promising design, El Rehab City faces challenges related to its reliance on private cars and the lack of pedestrian and cycling infrastructure, revealing an opportunity for transformation into a more sustainable, localized urban environment that could fully embody the 15-minute city concept.

The selection of Al-Rehab City as a model for studying the feasibility of implementing the '15-

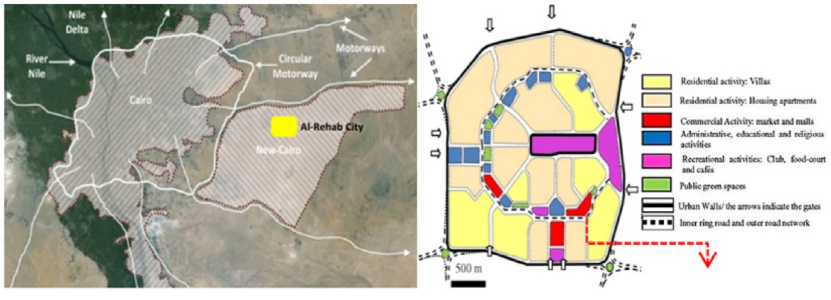


Fig. 2. Al-Rehab city location within New Cairo, the land use plan of the Western part of Al-Rehab city source (Staricco and Brovarone, 2022).

Table 1. Key features of El Rehab City.

Urban planning and design	Divided into ten phases, the city features a low-density layout with ample green spaces, parks, and pedestrian-friendly pathways, fostering a walkable environment.
Residential diversity	The city offers a variety of housing options, including apartments, villas, and townhouses, catering to different income levels and family needs, with architecture blending modern and traditional styles.
Comprehensive amenities	El Rehab includes schools, healthcare facilities, shopping centers, restaurants, sports clubs, and recreational facilities, all within easy reach. Public and private transportation links the city with broader New Cairo and Cairo
Sustainability measures	The city integrates green spaces and water features, coupled with efficient waste management and water conservation practices, contributing to environmental sustainability.
Security and community	With gated entrances and private security services, El Rehab ensures a high level of safety. Regular community events foster social interaction and a strong sense of belonging.
Strategic accessibility	Situated in New Cairo, El Rehab is well-connected via major roads and highways, such as the Cairo-Suez Road and the Ring Road, ensuring proximity to educational institutions, business districts, and Cairo International Airport.

Source (Yue *et al.*, 2024), edited by authors.



Fig. 3. Neighborhood service centers are in the middle of residential areas. Source (Staricco and Brovarone, 2022).

minute city' concept is based on several reasons. Firstly, the city follows the idea of residential neighborhoods in its planning, where services are concentrated in specific areas, like most new cities in Egypt. Additionally, as a gated community, it ensures the availability of many essential services.

The city also features a distinctive design with an emphasis on providing ample green spaces. Moreover, its size is suitable for implementing the 15-minute city concept, as the city spans an area of 10 km^2. This size allows for covering the distances from its center to its edges within 15 min on foot or by bicycle. This feature is a key factor in implementing the concept, like the Barcelona experience, where the city was divided into units known as 'superblocks,' and the concept was gradually applied in each area.

This makes Al-Rehab an ideal environment for applying the principles of livable cities and sustainable mobility within a limited and well-organized space. It also serves as a model that can be replicated in most new cities in Egypt, which follow a similar planning approach.

3.2. Comparative study between El Rehab City and Barcelona, Spain: applying the 15-minute city concept for sustainable urban development

The selection of El Rehab City in Egypt and Barcelona in Spain as case studies is driven by their distinct urban planning approaches and the opportunity to analyze the application of the 15-minute city model in diverse contexts. Barcelona is renowned for its innovative urban planning, particularly the Superblocks initiative, which has transformed the city into a more pedestrian-friendly environment. This initiative has garnered international attention for its success in reducing car dependency and enhancing urban livability. El Rehab City, developed by the Talaat Moustafa Group, represents a planned community in Egypt that aims to integrate residential, commercial, and recreational spaces. While it offers a comprehensive living environment, it faces challenges related to car dependency and limited pedestrian infrastructure. By examining these two cities, the research aims to understand how the 15-minute city model can be adapted and implemented in different cultural and infrastructural settings.

The 15-minute city concept emphasizes accessibility, sustainability, and improved quality of life by ensuring residents can meet their daily needs within a short walk or bike ride. This comparative study explores how this model can be implemented in El Rehab City, Egypt, and Barcelona, Spain, by analyzing their existing infrastructure, accessibility, and sustainability features, as shown as Table 2.

A comparative analysis of the potential for implementing the 15-minute city concept in El Rehab City, Egypt, and Barcelona, Spain, highlights

Table 2. Comparative study between El Rehab City and Barcelona, Spain.

	Barcelona, Spain	El Rehab City, Egypt
Existing infrastructure	Established urban fabric with mixed-use developments.	A planned community with defined residential zones and commercial areas.
	Extensive public transportation network, including buses, metro, and trams	Reliant on private cars for internal and external transportation
	Pedestrian-friendly areas, particularly through the Superblocks initiative	Limited pedestrian pathways and dedicated cycling infrastructure
	Integrated cycling lanes and bike-sharing systems	Gated community design restricts integration with surrounding areas
	Accessible public parks and green corridors connecting neighborhoods	Green spaces exist but are not interconnected
Accessibility	Services are well-distributed within Superblocks, with healthcare, education, and recreation accessible within 15 min walking distances	Essential services (e.g., schools, clinics, supermarkets) are concentrated but not evenly distributed
	Pedestrianized streets and low-traffic zones enhance accessibility.	Dependence on cars limits accessibility for nondrivers.
	Active mobility is promoted through safe, interconnected cycling networks	Walkability and bike ability hindered by the lack of shaded pathways and cycling lanes
Sustainability features	Pioneering green initiatives, including solar energy installations and waste recycling programs	Energy consumption primarily reliant on nonrenewable sources
	Urban greenery and eco-friendly transportation reduce carbon emissions	Insufficient integration of green energy and sustainable waste management
	The Superblocks model fosters community interaction, lowers air pollution, and reduces urban heat	Potential for solar energy utilization due to abundant sunlight

Source: by authors.

significant contrasts in their infrastructure, accessibility, and sustainability features. While El Rehab City benefits from its planned design, it relies heavily on private cars, lacks pedestrian and cycling infrastructure, and faces challenges in integrating services and green spaces. In contrast, Barcelona has a well-established urban fabric, supported by extensive public transportation, pedestrian-friendly Superblocks, interconnected cycling lanes, and accessible green corridors. Accessibility in El Rehab is hindered by unevenly distributed services and car dependence, whereas Barcelona excels with its walkable, service-rich neighborhoods. Regarding sustainability, El Rehab relies on non-renewable energy but has potential for solar energy utilization, while Barcelona leads with green initiatives like solar power, recycling programs, and eco-friendly transportation. This comparison underscores the need for tailored strategies to overcome El Rehab's challenges by drawing from Barcelona's innovative approaches to sustainable urban living.

A SWOT (Strengths, Weaknesses, Opportunities, and Threats) analysis of El Rehab City and Barcelona, Spain, reveals distinct opportunities and challenges in implementing the 15-minute city concept, as shown in Table 3. El Rehab City benefits from its planned design and potential for solar energy integration but faces challenges such as car dependency, gated community barriers, and limited pedestrian and cycling infrastructure. Opportunities include leveraging GIS for service optimization, developing shaded walkways, and utilizing green spaces, while threats include cultural car preferences, extreme summer heat, and financial constraints. Barcelona, on the other hand, showcases strengths like the successful Superblocks initiative, robust public transportation, and a strong political commitment to sustainability. However, its dense urban fabric, high maintenance costs, and resistance from car-dependent stakeholders pose challenges. Opportunities for Barcelona include expanding Superblocks, integrating smart technologies, and fostering participatory planning, while threats stem from economic pressures, resistance to change, and increasing demands from tourism. These insights highlight the tailored strategies required to adapt the 15-minute city concept to each context effectively.

Implementing the 15-minute city concept in El Rehab City requires targeted actions to improve accessibility, sustainability, and community interaction. This includes developing shaded pedestrian pathways, cycling lanes, and leveraging GIS to address service gaps. Sustainability measures should focus on integrating solar energy, initiating waste recycling programs, and expanding green spaces. To foster community engagement, underutilized areas can be repurposed into public plazas and community hubs, with cultural and recreational activities to enhance social cohesion. For Barcelona, the focus should be on scaling up the Superblocks initiative to underserved neighborhoods, incorporating digital tools to monitor mobility patterns, and promoting

Table 3. A SWOT analysis between El Rehab City and Barcelona, Spain.

	Barcelona, Spain	El Rehab City, Egypt
Strengths	Proven success of the Superblocks initiative	Planned urban design facilitates zoning for implementing the 15-minute city
	Established public transportation and cycling infrastructure	High potential for integrating solar energy due to the climate
	Strong political commitment to sustainability goals	Strong community identity within gated neighborhoods
Weaknesses	Dense urban fabric limits large-scale redevelopment	Heavy reliance on private cars limits accessibility and sustainability
	High cost of maintaining and expanding green initiatives	Gated community structure restricts connectivity
		Lack of pedestrian and cycling infrastructure
Opportunities	Continued expansion of Superblocks to other neighborhoods	Leverage GIS for mapping and optimizing service distribution
	Integration of smart city technologies for resource optimization	Develop shaded pedestrian walkways and cycling paths to encourage active mobility
	Enhanced community engagement through participatory urban planning	Utilize untapped green spaces for community hubs and recreational areas
Threats	Resistance from car-dependent residents and businesses	Cultural preference for private car use
	Economic challenges in scaling up sustainable infrastructure	High temperatures limiting outdoor mobility during summer months
	Rising tourism pressures increasing demand on urban services	Financial constraints for retrofitting infrastructure

Source: by authors.

green energy adoption through solar installations and energy-efficient practices. Strengthening public engagement through participatory planning and education on reduced car dependency will further enhance sustainability efforts. Together, these strategies highlight the adaptability of the 15-minute city model in diverse urban settings, with the potential to enhance accessibility, sustainability, and quality of life.

By analyzing their infrastructure, accessibility, and sustainability features, we aim to understand how each city reflects the eight key components of the 15-minute city model:

(a) Mixed-Use Development:

Barcelona has established a well-integrated urban fabric with mixed-use developments, allowing residents to access various services and amenities within their neighborhoods. In contrast, El Rehab City is a planned community with defined residential zones and commercial areas, which may limit the immediate availability of diverse services within walking distance.

(b) Proximity to Essential Services:

In Barcelona, essential services such as healthcare, education, and recreation are well-distributed within Superblocks, with most accessible within a 15-min walking distance. El Rehab City, however, has essential services concentrated but not evenly distributed, leading to potential accessibility challenges for residents.

(c) Sustainable mobility:

Barcelona promotes active mobility through safe, interconnected cycling networks and pedestrian-friendly streets, reducing reliance on cars. El Rehab City, on the other hand, is heavily reliant on private cars, with limited pedestrian pathways and dedicated cycling infrastructure, hindering sustainable mobility.

(d) Green spaces and public areas:

Barcelona integrates urban greenery and eco-friendly transportation to reduce carbon emissions, fostering community interaction and enhancing environmental sustainability. El Rehab City has green spaces, but they are not interconnected, and the community design restricts integration with surrounding areas, limiting the potential for communal green areas.

(e) Local economic development:

Barcelona's mixed-use developments support local businesses and economic activities within neighborhoods. El Rehab City's reliance on private cars and gated community design may limit local

economic interactions and opportunities for small businesses.

(f) Social inclusion:

Barcelona's Superblocks model fosters community interaction and inclusivity by creating shared spaces. El Rehab City's gated community structure restricts connectivity and may limit social interactions among residents.

(g) Health and well-being:

Barcelona's emphasis on walkability, cycling, and green spaces contributes to residents' physical and mental health. El Rehab City's car dependency and limited pedestrian infrastructure may hinder active lifestyles and access to recreational areas.

(h) Resilience and adaptability:

Barcelona's commitment to sustainability and green initiatives, such as solar energy installations and waste recycling programs, enhances its resilience. El Rehab City has potential for solar energy utilization due to abundant sunlight but currently lacks sufficient integration of green energy and sustainable waste management.

In conclusion, while Barcelona exemplifies many of the key components of the 15-minute city model through its integrated urban planning and sustainable practices, El Rehab City faces challenges in implementing these components due to its reliance on private cars, limited pedestrian and cycling infrastructure, and gated community design. Addressing these challenges in El Rehab City could involve developing shaded pedestrian walkways, cycling paths, and leveraging GIS for service optimization to enhance accessibility and sustainability.

This research aims to evaluate the application of the 15-minute city model in El Rehab City, Egypt, and Barcelona, Spain, by analyzing their infrastructure, accessibility, and sustainability features. It seeks to identify the strengths, weaknesses, opportunities, and threats associated with implementing this model in both cities. Additionally, the study proposes strategies to enhance the application of the 15-minute city model in El Rehab City, drawing insights from Barcelona's experiences.

The comparative study of El Rehab City and Barcelona provides valuable insights into the practical application of the 15-minute city model. Barcelona's success in creating pedestrian-friendly environments and reducing car dependency offers a framework that can be adapted to El Rehab City. Implementing strategies such as developing shaded pedestrian walkways, cycling paths, and leveraging GIS for service optimization can enhance accessibility and sustainability in El Rehab City. Additionally, integrating solar energy, initiating waste recycling programs, and expanding green spaces can contribute to environmental sustainability. Fostering community engagement through participatory planning and education on reduced car dependency will further enhance sustainability efforts. These tailored strategies highlight the adaptability of the 15-minute city model in diverse urban settings, with the potential to improve accessibility, sustainability, and quality of life.

3.3. Methodology for using GIS and network analysis to evaluate masterplan designs for a 15-minute city

Based on a review of previous literature, the study of Al-Rehab City, and its comparison with some global experiences, it was found that implementing the concept of '15-min cities' requires several essential elements. Among these, the most challenging in terms of implementation and cost is ensuring the availability of various services within a 15 min walking or cycling range. Accordingly, the practical aspect of the study focused on analyzing the availability of key services in the city and determining their zones of influence. To achieve this, GIS software was utilized to measure service accessibility through network analysis. To evaluate masterplan designs for a 15-Minute City using GIS and Network Analysis, a structured methodology can be employed. This methodology involves several steps, from data collection to analysis and evaluation. Here's a structured approach.

3.3.1. Defining the 15-min distance criteria

(a) Time-distance thresholds: Define what constitutes a 15 min walk, bike ride, or transit trip. Typically, this is about 1.2 km for walking and 4.8 km for biking.
(b) Service accessibility: Determine which essential services should be accessible within these time thresholds.

Due to the variation in distances covered in 15 min by walking or cycling, the types and levels of services that should be available within each distance have been classified accordingly. The essential services that must be present in 15-min cities as shown as Table 4.

Table 4. Main services within 15-minute.

	On foot (1.200 km)	By bike (4.828 km
Commercial services	Supermarket, Bakery, Grocery …	Malls and commercial centers (retail stores)
Entertainment services	Green and natural areas	Parks, sports, and entertainment clubs
Religious services	Mosques	Churches, Mosques
Educational services	Nursery, Primary school and Primary school	Secondary schools and Higher education
Public transportation	Ride points, bus station	Ride stations, Metro
Medical services	Clinics-Pharmacies	hospital, and urgent care
Administrative services	Banks, postal centers	Police and fire stations, Municipal offices, civil services offices.
Civic centers	_____	City Hall, libraries, Justice Center, and Museum

Source: by authors.

3.3.2. Methodology steps for analyzing and applying the 15-minute city concept

Based on the previous classification of services within a 15-minute city, a geographic database was created for Al-Rehab City based on satellite images available online through the program, in addition to collecting the required information through field surveys done by the authors. This database includes the road network and a base map containing essential information on land use and service locations. The services were categorized into those located within a 1.2 km range (15 min on foot) and those that should be available within a 4.8 km range (15 min by bike), Building on the geographic database developed for Al-Rehab City, which maps road networks, land use, and service locations based on walking and cycling accessibility, the subsequent methodology outlines the steps for analyzing and applying the 15-minute city concept. This methodology can be illustrated through the following diagram, Fig. 4.

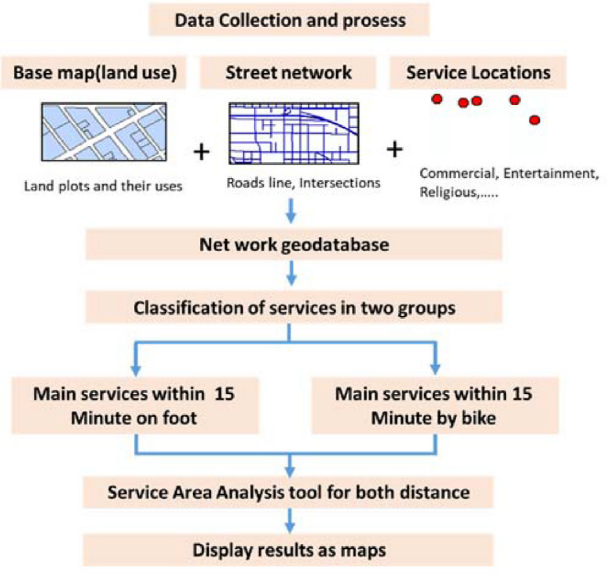


Fig. 4. Methodology for using GIS and network analysis to evaluate a 15-minute city. Source: by authors.

3.3.3. Data collection and organizing

(a) Base maps and masterplans: Acquire base maps and masterplan designs, including land use, transportation networks, and locations of key amenities (e.g., schools, healthcare, retail, parks).
(b) GIS data layers: Collect GIS layers such as road networks, pedestrian pathways, cycling routes, and public transit routes, ensuring accurate georeferencing.
(c) Population and demographic data: Collect population distribution and demographic data to understand the needs of various community segments.
(d) Service locations: Identify and map the locations of essential services, including educational institutions, healthcare facilities, grocery stores, parks, and recreational centers

3.3.4. Analysis and operations

Network analysis evaluates the connectivity of urban locations. It assesses transportation networks, including roads and pathways, to determine how easily residents can reach essential services. Key functions include (Khavarian-Garmsir *et al.*, 2023a).

(a) Network creation: Create a transportation network in GIS that includes roads, pedestrian paths, cycling routes, and public transit lines. Ensure that the network topology is correct, with nodes representing intersections and edges representing pathways.

Service area analysis: Use network analysis tools (service area analysis) to generate service areas around each key amenity. These polygons represent the area within which residents can access services within 15 min. This analysis will be conducted in two stages for all services: the first stage within a 15 min walking distance, equivalent to 1.2 km, and the second stage within a 15 min biking distance, equivalent to 4.828 km.

3.3.5. Scenario testing and comparison

GIS and network analysis allow urban planners to model various development scenarios and assess their impacts on accessibility. For instance, planners can simulate the effects of adding new parks or bike lanes, optimizing service placement to align with the 15-minute city framework (Carpentieri *et al.*, 2019).

(a) Masterplan evaluation: Apply network analysis to masterplan design, assessing how well meets the 15 min accessibility criteria.
(b) Gap identification: Highlight areas of insufficient accessibility or service deficiencies to inform future modifications.

3.3.6. Visualization and interpretation

(a) GIS mapping: Create thematic maps to visualize accessibility levels, service coverage gaps, and potential barriers (e.g. major roads).
(b) Network diagrams: Create diagrams illustrating connectivity and accessibility among services, aiding in understanding spatial relationships flow within the urban fabric.

This methodology leverages GIS and network analysis to systematically assess masterplan designs for a 15-minute city, ensuring that urban environments are designed to be accessible, equitable, and sustainable. By following these steps, planners can enhance the quality of life for residents and promote localized urban living.

4. Results and discussion

A network analysis was conducted on the database of Al-Rehab City using the Service Area Analysis tool in ARC GIS 10.5. The analysis was performed in two stages: first, for the 15 min walking range, which typically includes services found within residential neighborhoods, and second, for the 15 min biking range, covering services at the district level. This methodology allowed for a comprehensive understanding of service accessibility across different scales of the city. The following section presents a detailed analysis of the results for each service category.

4.1. First: commercial services

The locations of commercial services within Al Rehab City have been systematically identified, with services distributed across designated commercial centers. The analysis of service coverage was conducted in two distinct phases. The first phase focused on identifying areas within a 15 min walking distance (1.2 km), covering services such as grocery stores and supermarkets. The second phase expanded the analysis to a 15 min cycling distance (4.8 km), encompassing services such as hypermarkets and shopping malls. The results of this analysis, which detail the areas covered by commercial services in both phases, are shown in Fig. 5.

The analysis revealed that the areas covered by commercial services within 15 min walking distance

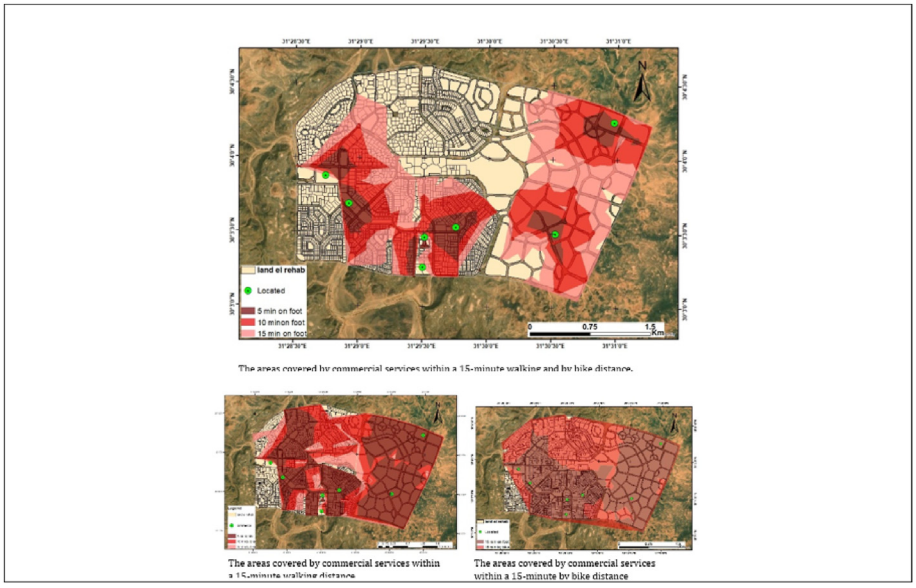


Fig. 5. The areas covered by commercial services within a 15 min walking and by bike distance. Source: by authors using ARC GIS 10.5 program.

span 5.8 km^2, representing ~58 % of Al Rehab City's total area. In contrast, when considering a 15 min cycling distance, the coverage extends to 9.7 km^2, which accounts for about 98 % of the city's total area.

4.2. Second: entertainment services

The locations of entertainment services within Al Rehab City have been systematically identified, with these services regularly distributed across designated centers. The analysis of service areas was carried out in two phases. The first phase focused on identifying areas covered by entertainment services within a 15 min walking distance (1.2 km), including services such as small parks and green spaces. The second phase examined areas covered by entertainment services within a 15 min cycling distance (4.8 km), encompassing services such as large parks and sports clubs. Fig. 6 below illustrates the results of the analysis, highlighting the areas covered by entertainment services in both phases.

4.3. Third: religious services

Al-Rehab City contains eight mosques distributed across its area and one church located in the western part of the city. The analysis of religious service coverage was conducted in two phases: first, for mosque services, by assessing both 15 min walking and cycling distances, and second, for the church service, by considering a 15 min cycling distance only. As shown in Fig. 7, the coverage of religious services for mosques within a 15 min walk spans 5 km^2, which constitutes approximately 51 % of the city's total area. In contrast, the areas covered by mosque services within a 15 min bicycle ride extend to 8.7 km^2, representing about 88 % of the city's area. Fig. 8 illustrates the areas covered by religious services for the church, which spans an area of 4.5 km^2, representing ~46 % of the city's total area.

4.4. Fourth: educational services

Educational services within Al-Rehab City are limited to four private schools offering various

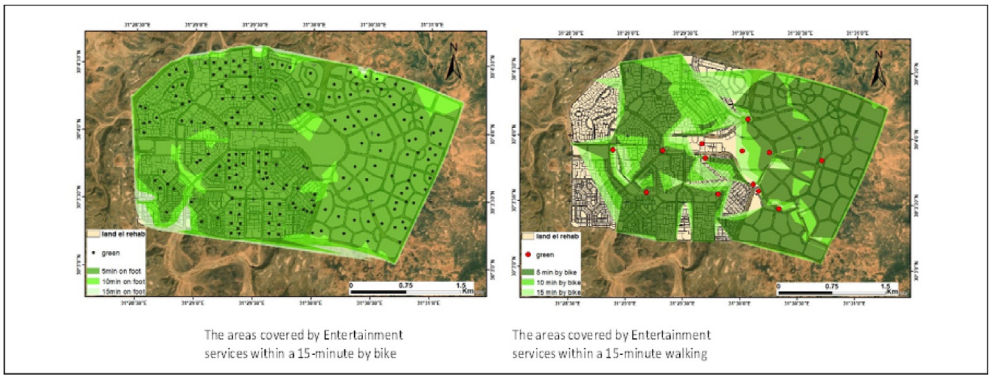


Fig. 6. The areas covered by Entertainment services within a 15 min walking and by bike distance. Source: by authors using ARC GIS 10.5 program. The analysis revealed that the areas covered by entertainment services within 15 min walking distance span 9.8 square kilometers, representing ~99 % of Al Rehab City's total area. In contrast, the areas covered by entertainment services within a 15 min cycling distance extend to 8.3 km^2, which accounts for about 84 % of the city's area.

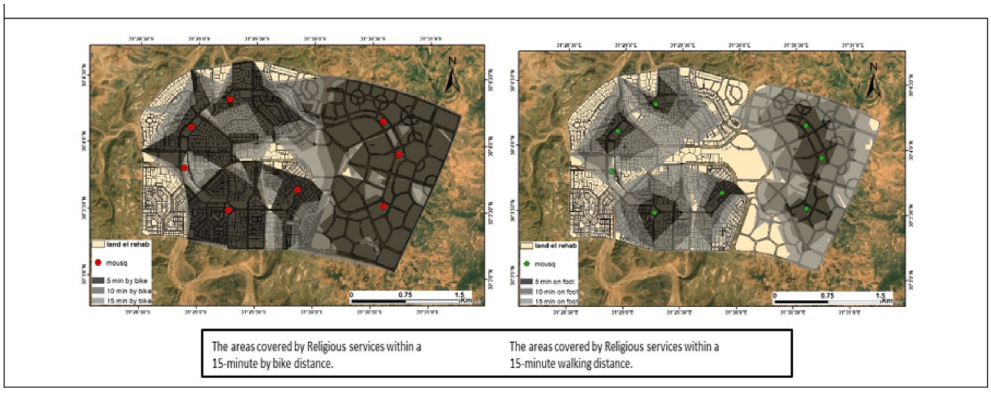


Fig. 7. The areas covered by religious services within a 15 min walking and by bike distance. Source: by authors using ARC GIS 10.5 program.

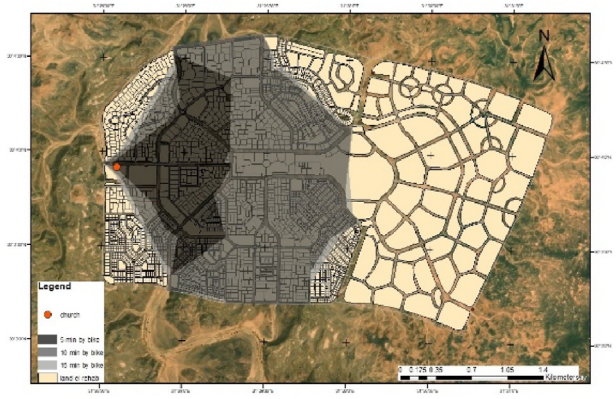


Fig. 8. The areas covered by Religious (church) services within a 15 min by bike distance. Source: by authors using ARC GIS 10.5 program.

educational levels, with no higher education institutions available. Consequently, the coverage area for educational services was analyzed based solely on a 15 min walking distance, given the absence of higher-level educational facilities. The analysis reveals that the areas covered by educational services span 5.4 km^2, accounting for approximately 54 % of the city's total area, as shown in Fig. 9.

4.5. Fifth: public transportation

Transportation services in this context refer to access points for public transportation, such as bus stations and bus stop locations. The coverage of transportation services within a 15-minute city is illustrated in Fig. 10. The analysis shows that areas covered by transportation services within 15 min walking distance span 2.2 square kilometers, representing ~23 % of the city's total area. In contrast, the areas accessible within a 15 min bike ride encompass 9.97 km^2, covering nearly 100 % of the city's area.

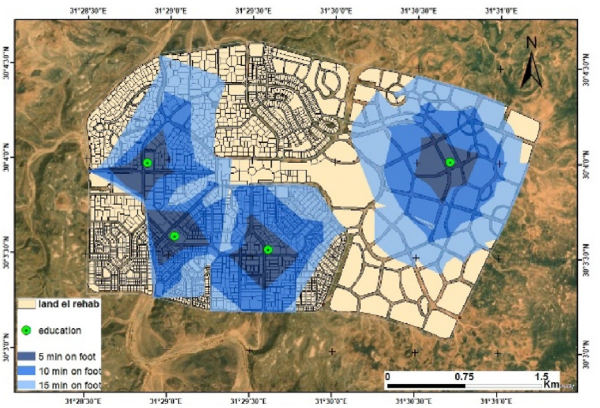


Fig. 9. The areas covered by educational services within a 15 min by bike distance. Source: by authors using ARC GIS 10.5 program.

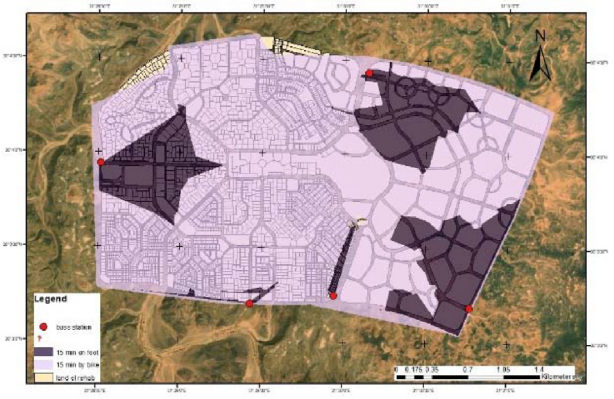


Fig. 10. The areas covered by Transportation services within a 15 min walking distance. Source: by authors using ARC GIS 10.5 program.

4.6. Sixth: medical services

Medical services within Al-Rehab City are limited to clinics and pharmacies. To assess the accessibility of these services, the service area was calculated based on a 1.2 km radius, corresponding to a 15 min walking distance. The area covered by medical services within this range is 1.9 km^2, which represents ~19 % of the city's total area, as depicted in Fig. 11. However, higher-level medical services, such as hospitals or specialized medical centers, are not available within the city.

4.7. Seventh and finally: administrative services

The distribution of administrative services within the city was analyzed in two phases. The first phase aimed to identify areas covered by administrative services within a 15 min walking distance (1.2 km), encompassing services such as banks and postal centers. The second phase focused on identifying areas accessible within a 15 min cycling distance

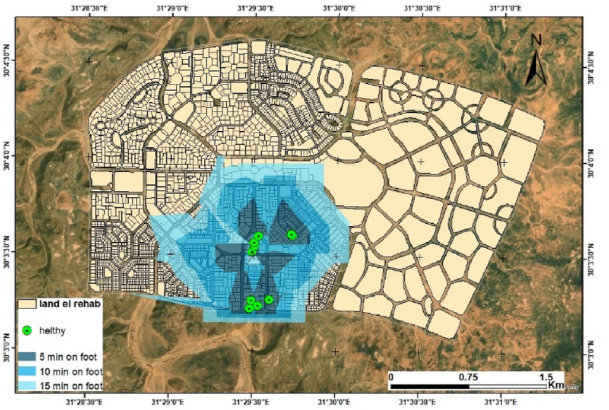


Fig. 11. The areas covered by medical services within a 15 min walking distance. Source: by authors using ARC GIS 10.5 program.

(4.8 km), which includes services such as police and fire stations, municipal offices, and civil service offices. The results of this analysis, illustrated in Fig. 8, depict the spatial coverage of administrative services in both phases.

As shown in Fig. 12, the coverage of administrative services within a 15 min walking radius spans an area of 2.5 km^2, accounting for ~25 % of the city's total area. In contrast, the areas covered by administrative services within a 15 min cycling radius extend to 8.9 km^2, encompassing about 90 % of the city's area. This analysis highlights the significant difference in service accessibility when comparing walking and cycling distances, underscoring the broader coverage achievable through cycling as a mode of transportation.

The analysis of service coverage in El Rehab City, as detailed in Table 5, highlights significant disparities in the accessibility of various services, providing insights that can inform urban planning across both existing and new cities in Egypt. Recreational services exhibit near-optimal spatial coverage, with 99 % of the city accessible within a 15-min walking distance, attributed to the integration of green spaces across residential clusters. However, critical services, such as medical and educational facilities, reveal substantial deficits. Medical services are concentrated in the city center, covering only 19 % of the area within walking distance, while educational services cover 54 %, reflecting a misalignment in service prioritization. The clustering of commercial areas, mosques, and educational institutions into small hubs is noteworthy, yet only 50 % of the population can access these hubs on foot. This raises concerns about public health and urban equity, as the current urban design favors recreational accessibility over the equitable distribution of essential services.

Table 5. The ratio of the area covered by various services to the total area of Al-Rehab City.

Services	On foot, %	By bike, %
Commercial	**58**	**98**
Entertainment	**99**	**84**
Religious		
Mosques	**51**	**88**
Church	–	**46**
Educational	**54**	–
Public transportation	**23**	**100**
Medical	**19**	–
Administrative	**25**	**90**
Civic centers	–	–

Source: by authors.

To address these challenges and align with Egypt's Vision 2030, lessons from successful global 15-minute city models, such as those in Paris, Ottawa, and Amsterdam, can be adapted to the country's unique socioeconomic and spatial realities. For existing informal settlements, strategies should focus on upgrading infrastructure through affordable mixed-use zones, community-driven planning, and the introduction of mobile clinics, shared transit hubs, and pedestrian pathways to improve accessibility. In new cities, such as El Rehab, decentralizing essential services by establishing satellite facilities in underserved areas and fostering compact, connected urban hubs can

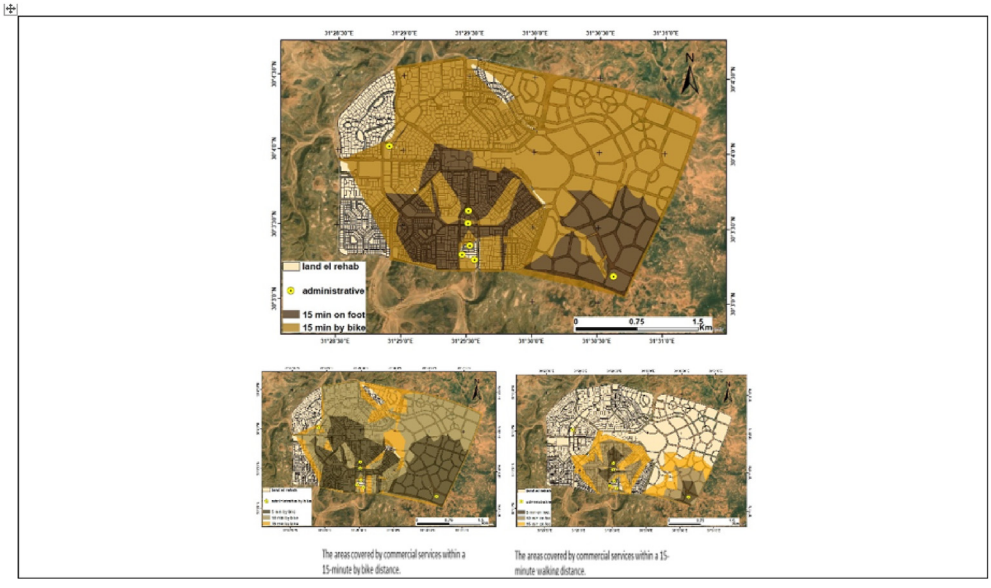


Fig. 12. The areas covered by administrative services within a 15 min distance. Source: by authors using ARC GIS 10.5 program.

enhance spatial equity. Public-private partnerships (PPPs) can play a pivotal role in financing infrastructure development for healthcare, education, and transportation, accelerating implementation while maintaining quality standards.

Moreover, revising land-use policies to incorporate mixed-use zoning and leveraging GIS-based spatial analysis can support evidence-based resource allocation, ensuring that service distribution aligns with the needs of diverse communities. Benchmarking against global models can also inform the design of localized solutions that prioritize affordability, inclusivity, and sustainability. By integrating these approaches into national urban planning frameworks, Egypt can bridge accessibility gaps in both existing and planned cities, advancing the principles of the 15-minute city and achieving the goals outlined in Vision 2030. This research provides a replicable framework for creating equitable, sustainable urban environments across the country.

4.8. Conclusion

The 15-minute city model presents a compelling framework for fostering sustainable and equitable urban development, particularly within the context of El Rehab City, Egypt. This research highlights the critical role of spatial planning in advancing this model, revealing a stark contrast between the availability of recreational services and green spaces versus the limited access to essential services such as healthcare and education. While El Rehab City, characterized as a middle-to-upper-class gated community, benefits from well-structured service coverage, the challenge lies in adapting this model to Egypt's diverse urban landscapes, especially in informal settlements and low-income areas. The sprawling nature of cities like Cairo complicates the implementation of the 15-minute city concept due to existing infrastructure deficits and unequal service distribution.

Utilizing GIS and network analysis, this study identifies opportunities for creating decentralized service hubs, deploying mobile healthcare units, and promoting mixed-use zoning to enhance accessibility in underserved regions. Furthermore, it underscores the necessity for a holistic approach to urban planning that prioritizes improvements in public transportation, pedestrian infrastructure, and the integration of 15-minute city principles into national policies such as Egypt's Vision 2030. Successfully adapting this model requires addressing the unique socio-economic and spatial challenges faced by marginalized communities.

Future research should emphasize expanding GIS applications in urban planning, particularly for cities with inadequate infrastructure. This includes developing detailed GIS databases to better understand accessibility issues and target infrastructure improvements effectively. Additionally, cultural and climatic factors must be considered when designing urban mobility solutions, particularly in regions like the Middle East where extreme heat poses unique challenges. Engaging local communities in the planning process is essential to ensure that urban interventions align with the diverse needs of residents.

In summary, this study contributes valuable insights into applying the 15-minute city model within Egypt and similar contexts. It provides a scientific framework for developing urban environments that are sustainable, inclusive, resilient, and responsive to the needs of all residents. By leveraging GIS technology and prioritizing social equity in urban planning practices, cities can evolve into spaces that enhance the quality of life for their inhabitants while addressing contemporary urban challenges.

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Ethical statement

None.

Authors' contributions

All authors read and approved the final manuscript.

Conflicts of interest

There are no conflicts of interest.

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